

IDX Exchange DS34

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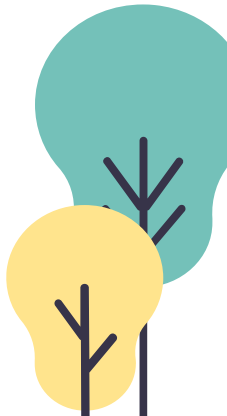
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Problem Statement

The monthly data is collected from Trestle. It contains historical close price data on real estate properties sourced from CRMLS (California Regional Multiple Listing Service), including various features such as living area, number of bedrooms, bathrooms, lot size, and close prices.



Real Estate



Data Cleaning

Original Data

- 178,507 rows and 80 columns
- Many columns with missing values

Steps taken

- Remove unusable or extremely incomplete data
- Restrict to comparable residential properties
- Price outlier removal
- Convert date fields
- Fill important structural features
- Geographic repairs
- Boolean feature normalization
- Final numeric cleanup

Final Data

- Training: 88,920 rows and 23 columns, February-September
- Testing: 4,864 rows and 23 columns, October

Data Cleaning

	BuyerAgentAOR	ListAgentAOR	Flooring	ViewYN	WaterfrontYN	BasementYN	PoolPrivateYN	OriginalListPrice	ListingKey	ListAgentEmail	...	M
0	RanchoSoutheast	RanchoSoutheast	NaN	True	NaN	NaN	NaN	60000.0	526199946	cmark1018@yahoo.com	...	
1	InlandValleys	InlandValleys	NaN	False	NaN	NaN	NaN	550000.0	525585060	mozcorona@aol.com	...	
2	SanDiego	SanDiego	NaN	False	NaN	NaN	False	880000.0	497696903	lenskab@gmail.com	...	
3	SanDiego	SanDiego	NaN	False	NaN	NaN	False	875000.0	497696407	lenskab@gmail.com	...	
4	SanDiego	SanDiego	NaN	False	NaN	NaN	False	849000.0	486616176	lenskab@gmail.com	...	

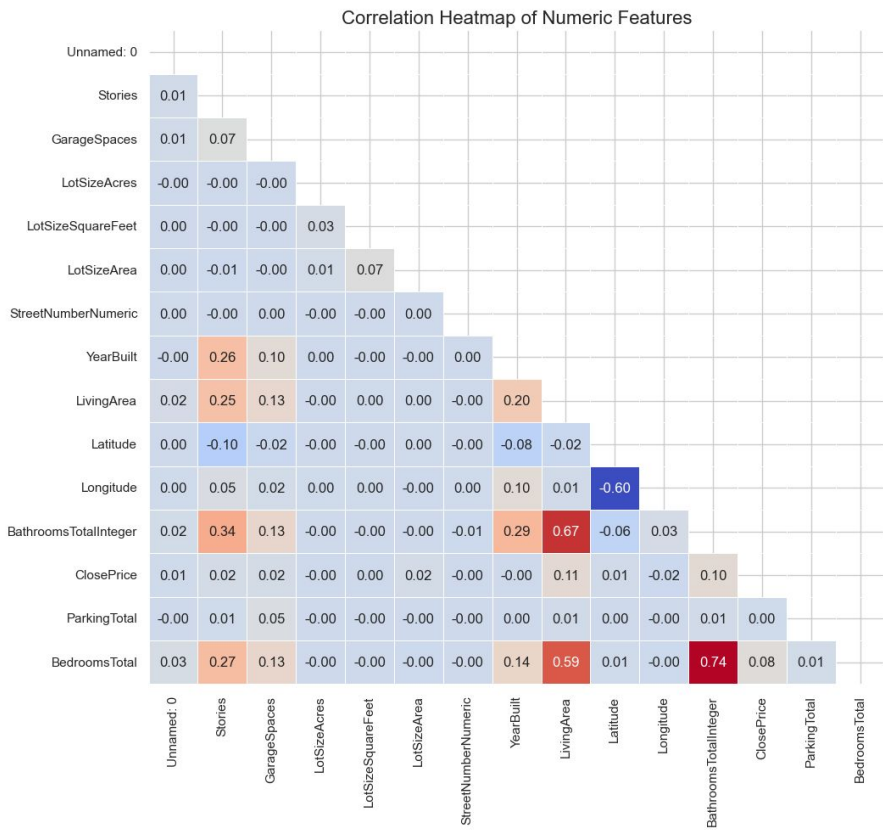
	ViewYN	PoolPrivateYN	CloseDate	ClosePrice	Latitude	Longitude	LivingArea	CountyOrParish	AttachedGarageYN	ParkingTotal	...	BedroomsTotal	C
2	False	False	2025-02-11	875000.0	32.765380	-117.043486	2340.0	San Diego	True	6.0	...	4.0	
3	False	False	2025-02-14	875000.0	32.765038	-117.043568	2165.0	San Diego	True	4.0	...	5.0	
4	False	False	2025-02-18	849000.0	32.765031	-117.043252	2158.0	San Diego	True	4.0	...	4.0	
15	True	False	2025-02-28	1100000.0	33.724000	-118.294924	2545.0	Los Angeles	False	3.0	...	4.0	
16	False	False	2025-02-28	760000.0	37.687556	-122.150114	1692.0	Alameda	True	2.0	...	4.0	



Data Cleaning

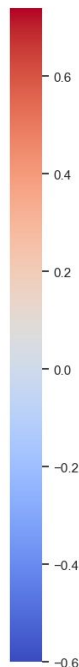
```
Index(['ViewYN', 'PoolPrivateYN', 'CloseDate', 'ClosePrice', 'Latitude',  
      'Longitude', 'LivingArea', 'CountyOrParish', 'AttachedGarageYN',  
      'ParkingTotal', 'YearBuilt', 'BathroomsTotalInteger', 'City',  
      'BedroomsTotal', 'ContractStatusChangeDate', 'PurchaseContractDate',  
      'ListingContractDate', 'FireplaceYN', 'Stories', 'NewConstructionYN',  
      'GarageSpaces', 'PostalCode', 'PropertyAgeAtClose'],  
      dtype='object')
```

Exploratory Data Analysis



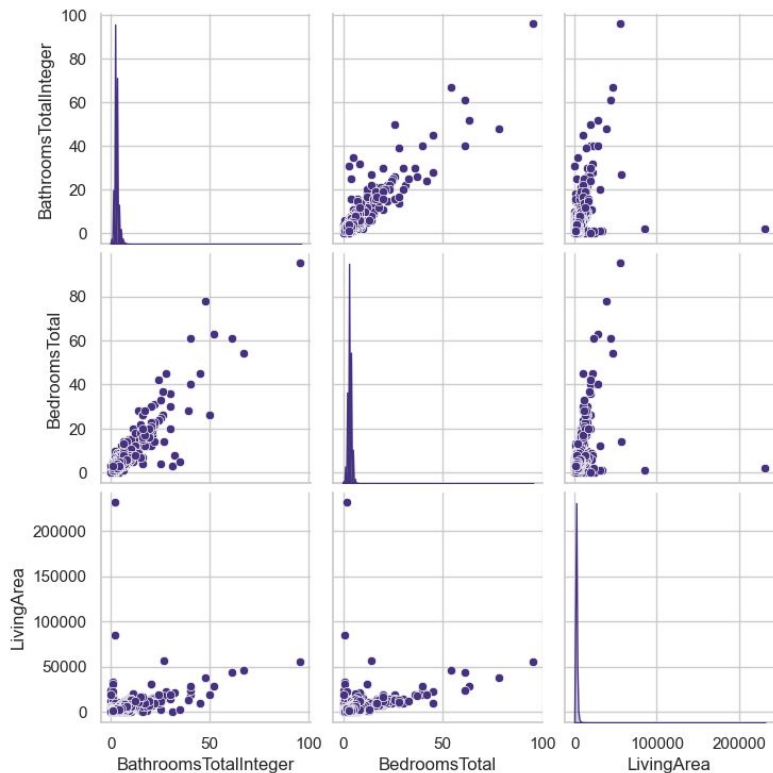
CORRELATION HEATMAP

We found strong correlations, especially between living area, bathrooms, and bedrooms.



Exploratory Data Analysis

Pairwise Relationships Among Top Correlated Features



PAIRWISE SCATTERPLOTS

To dig deeper, we plotted the top 5 correlated feature pairs, confirming linear clustering patterns. This gave us the confidence to create ratio and interaction-based engineered features later.



Models



Ridge/Lasso Model

Two versions of linear regression that include a regularization term that controls size of model's coefficients. They handle noisy data better, improve model robustness, and create simpler models.

Add penalties to model coefficients

Ridge: L1 penalty
(squared magnitude of coefficients)

Lasso: L2 penalty
(absolute value of coefficients)

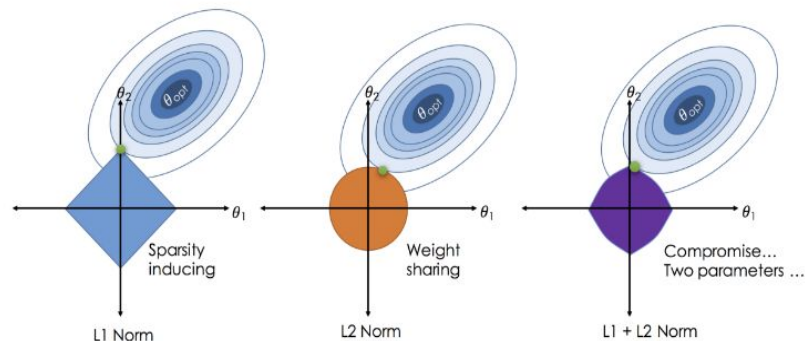
Control Model Complexity

Both methods prevent coefficients from becoming large, making model simpler and stabler.

Impact coefficients differently

Ridge: shrinks coefficient continuously

Lasso: shrinks coefficient exactly to zero



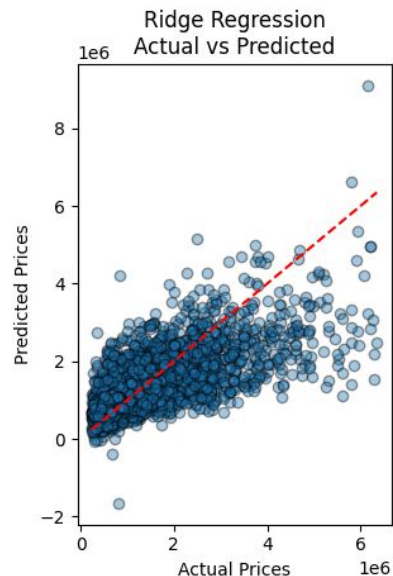
Ridge & Lasso: Results & Interpretation

Key Takeaways:

- Both models perform similarly, meaning underlying relationships are mostly linear and regularization doesn't drastically change results
- ~49% of price variance explained
- Errors remain 28 - 40%, indicating that linear models struggle with complex housing dynamics

Conclusion:

Ridge/Lasso provides a solid linear baseline, but price prediction likely requires nonlinear models and richer feature engineering to meaningfully improve accuracy.

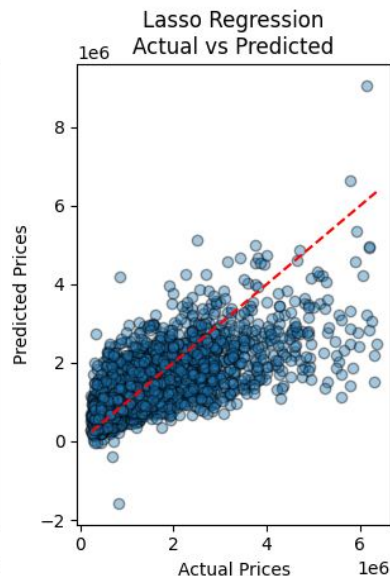


Ridge:

R^2 : 0.4896

MAPE: 40.84

MdAPE: 28.18



Lasso

R^2 : 0.4892

MAPE: 40.83

MdAPE: 28.23

XGBoost Model

Extreme gradient boosting is a high-performance machine learning algorithm based on boosted decision trees. It builds many small decision trees sequentially, where each new tree corrects the errors of the previous ones.

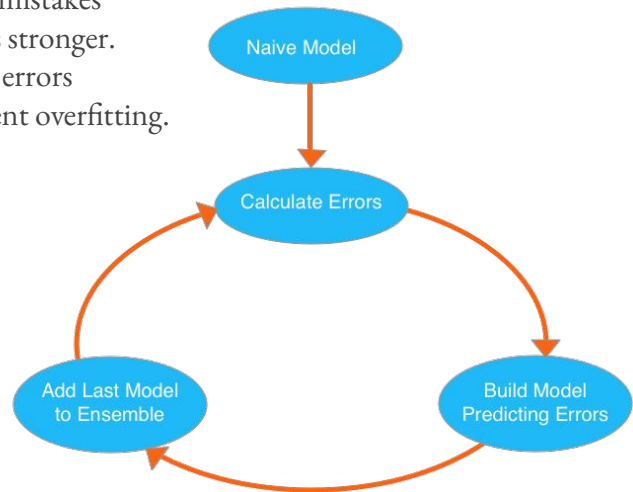
How it works

1. Start with a weak model (like a small tree)
2. Measure mistakes the model is making
3. Build a new tree focused on correcting those mistakes
4. Repeat hundreds of times → model becomes stronger.
5. Uses gradient descent to optimize prediction errors
6. Applies regularization and shrinkage to prevent overfitting.

Learns complex,
nonlinear relationships

Captures interactions
between features
automatically

Designed for tabular data
with mixed feature types



XGBoost

Temporal Features

- close_month
- close_quarter
- close_day_of_week
- days_contract_to_close

Location Features

- Latitude
- Longitude
- CountyOrParish_encoded
- City_encoded
- PostalCode_encoded

Home Amenities

- ViewYN
- PoolPrivateYN
- FireplaceYN
- AttachedGarageYN

Structural / Property Features

- LivingArea
- Stories
- BedroomsTotal
- BathroomsTotalInteger
- GarageSpaces
- ParkingTotal
- YearBuilt
- PropertyAgeAtClose
- NewConstructionYN

```
model = xgb.XGBRegressor(  
    n_estimators=100,      # Number of trees  
    learning_rate=0.1,    # Step size for each tree  
    max_depth=6,          # Maximum depth of each tree  
    min_child_weight=1,   # Minimum samples in leaf  
    subsample=0.8,        # Fraction of samples for each tree  
    colsample_bytree=0.8, # Fraction of features for each tree  
    random_state=42,  
    n_jobs=-1  
)
```

XGBoost- Results

TRAINING SET Metrics:

R^2 : 0.8779

MAPE: 16.35%

MdAPE: 11.65%

TEST SET Metrics:

R^2 : 0.8488

MAPE: 16.98%

MdAPE: 11.96%

OVERFITTING CHECK:

R^2 difference: 0.0291

✓ Model generalizes well

Interpretation: **Solid baseline performance**

Limitations + Areas for improvement

- Treating categorical Zip Codes as integers isn't ideal
- No external location features: demographics, school scores, proximity to landmarks, etc.
- No hyperparameter tuning (Use GridSearchCV)

Plans for improvement:

- Engineer more features: Neighborhood-level aggregation, such as mean price by postal code, rolling 90-day avg. price
- Hyperparameter tuning

Random Forest

WHY FEATURE ENGINEERING IS IMPORTANT FOR RANDOM FOREST?

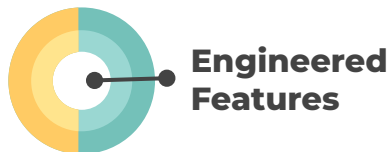
**Multiple
Listing
Service data
is sparse and
inconsistent**

**Many
predictive
relationships
are not linear**

**Goal: extract
structural +
temporal +
spatial
signals**



Random Forest Model



Price per SqFt

Bathrooms per Bedroom

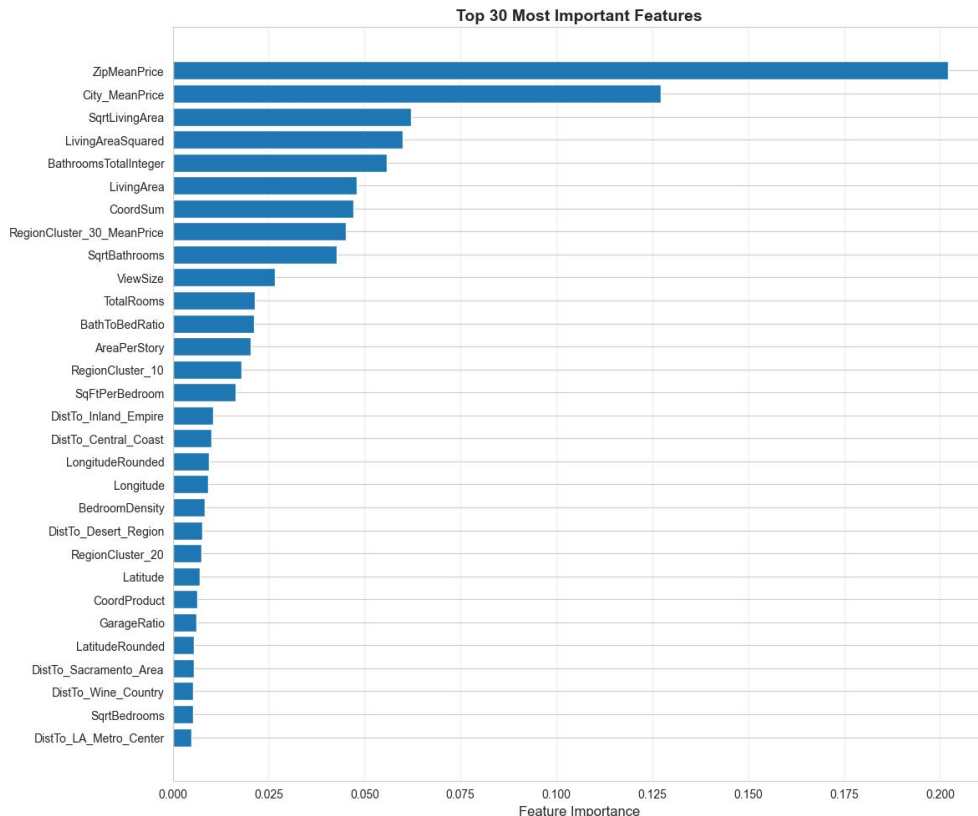
LotSize/LivingArea ratio

Age of property

Day, Month, Quarter of
listing/closing

County-level aggregates
(means/stats)

Binary flags for missing value
patterns



Random Forest Model

Learns complex,
nonlinear relationships

Not sensitive to scaling

Robust to noise and missing
values

How it works

1. Ensemble model made up of many decision trees
2. Each tree learns a slightly different view of the data
3. Final prediction = average of all trees
4. Reduces overfitting and captures nonlinear patterns
5. Works well with mixed data types and messy features

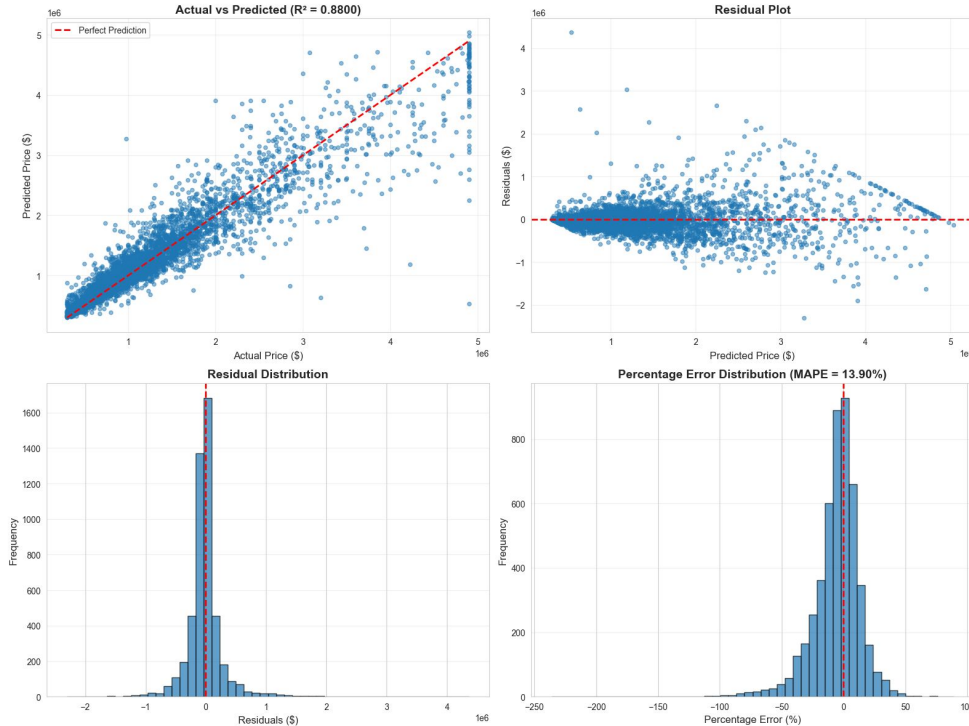
Performing 5-fold cross-validation...

Fold 1: $R^2=0.8844$, RMSE=304192.88, MAPE=13.77%
Fold 2: $R^2=0.8837$, RMSE=304129.99, MAPE=13.65%
Fold 3: $R^2=0.8840$, RMSE=304098.76, MAPE=13.87%
Fold 4: $R^2=0.8868$, RMSE=306527.51, MAPE=13.78%
Fold 5: $R^2=0.8780$, RMSE=314906.67, MAPE=13.83%

Cross-Validation Results:

	fold	MSE	RMSE	R^2	MAPE
0	1	9.253331e+10	304192.878001	0.884439	13.768924
1	2	9.249505e+10	304129.988663	0.883720	13.652708
2	3	9.247605e+10	304098.758551	0.884015	13.865835
3	4	9.395911e+10	306527.507935	0.886772	13.778667
4	5	9.916621e+10	314906.665528	0.878043	13.827299

Random Forest- Results



Interpretation: **Competitive performance**

Limitations + Areas for improvement

- Random Forests struggle with extrapolation
- Harder to interpret compared to linear models
- Training cost increases with feature count
- Sensitive to the quality of engineered features

TEST SET RESULTS

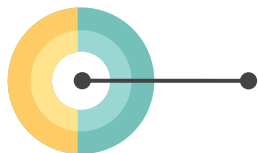
R^2 Score: 0.8800
MSE: 94,280,127,716.43
RMSE: 307,050.69
MAPE: 13.90%

Comparison to Original Random Forest Model:

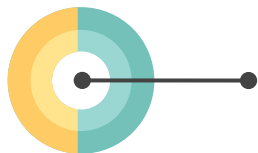
Original R^2 : 0.745
Improved R^2 : 0.8800
Improvement: 18.13%

Original RMSE: 463,173.73
Improved RMSE: 307,050.69
Improvement: 33.71%

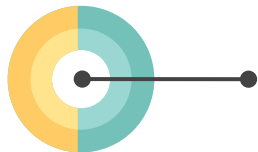
LightGBM (Light Gradient-Boosting Machine)



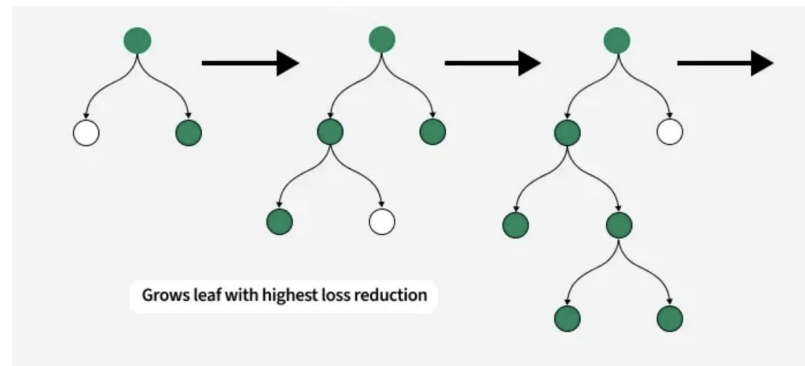
**LEAF-WISE
GROWTH STRATEGY**



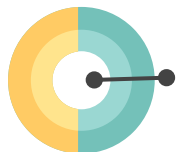
MEMORY + SPEED:
Histogram Binning and
Exclusive Feature
Bundling



TRAINING EFFICIENCY:
Gradient-Based One-Side
Sampling (GOSS)



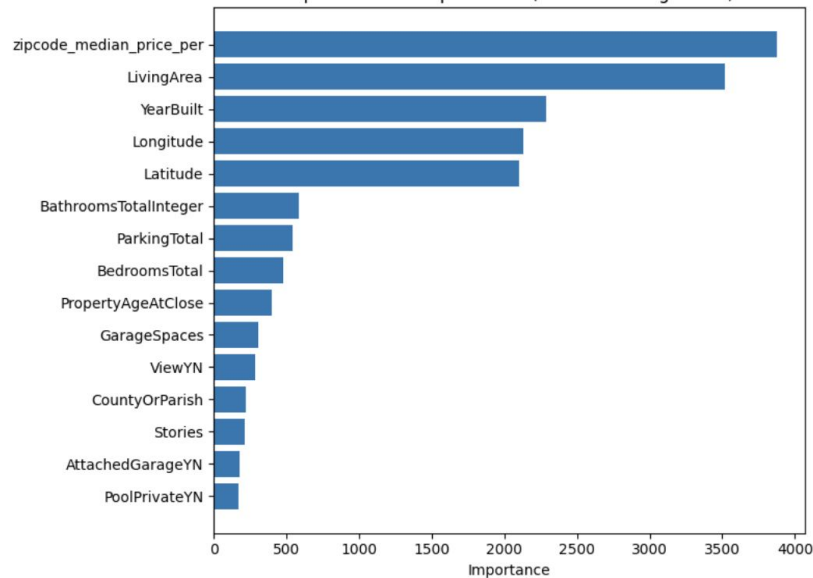
LightGBM (Light Gradient-Boosting Machine)



Development Process

1. **Featured Engineering**
2. **Data Pipeline**
3. **Created and Tested Model**
 - a. Original LightGBM Model
4. **Adjustments**
 - a. Feature Importance Cutoff / Threshold
 - b. Reduced LightGBM Model
5. **Tested Model**

Top 15 Feature Importances (Full-Feature LightGBM)



LightGBM (Light Gradient-Boosting Machine)



Evaluation/Performance

Reduced LightGBM (Full Training)

Reduced LightGBM (Full Testing)

Reduced LightGBM (Oct Testing)

MAPE

7.094309

14.777230

15.242247

R²

0.988304

0.871841

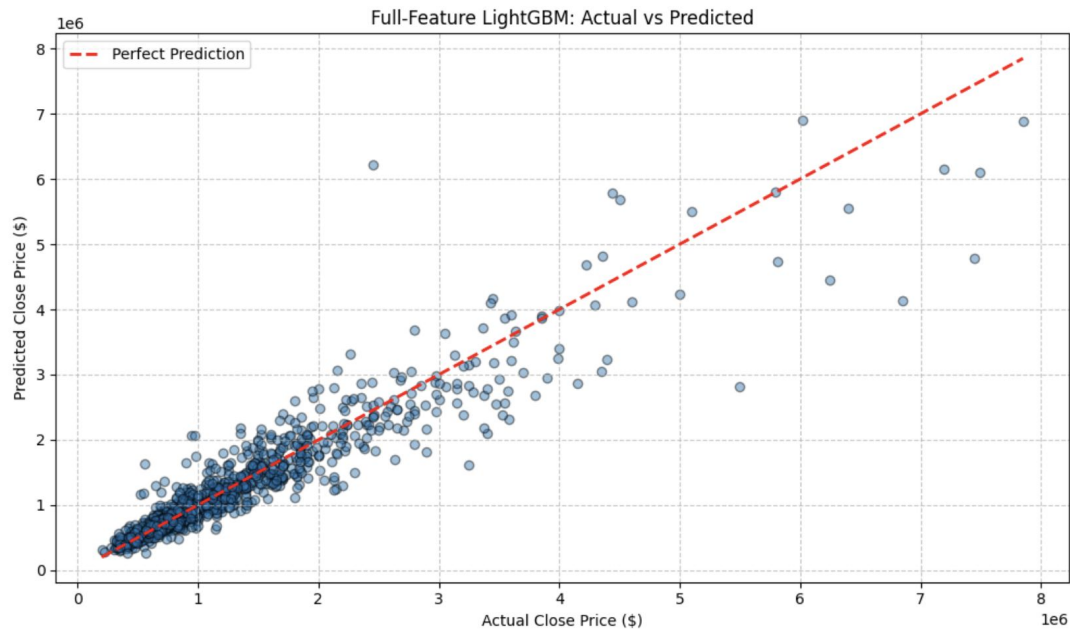
0.847924

MdAPE

5.080679

11.053140

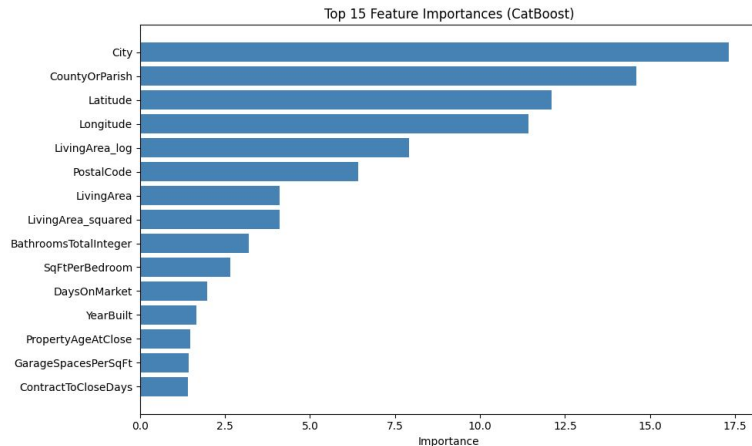
11.358144



CatBoost

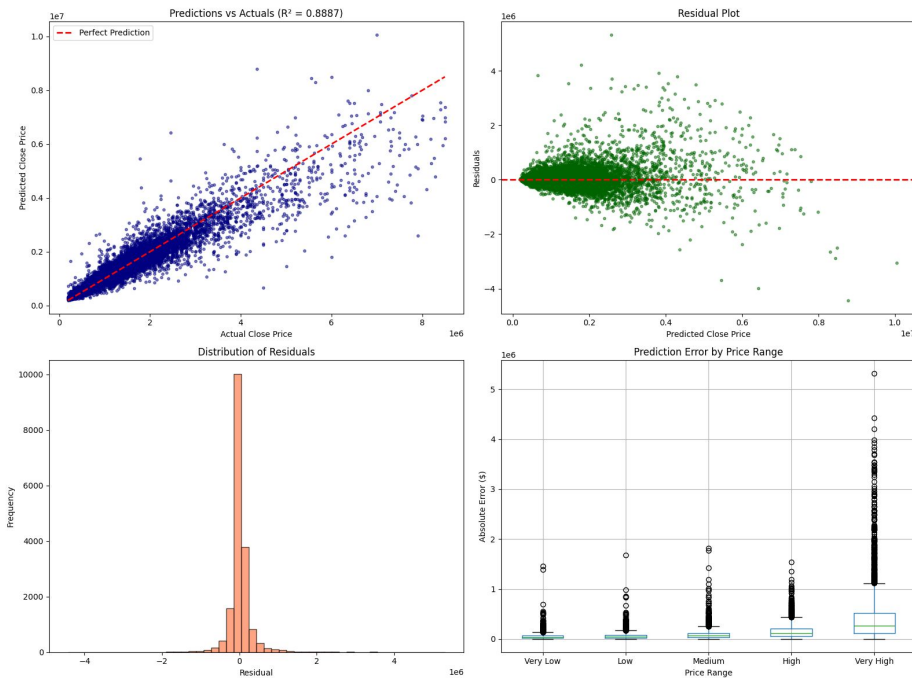
CatBoost is a gradient boosting algorithm that builds sequential decision trees to make predictions. We chose it because:

- Handles categorical data natively - our dataset has 900+ cities, which CatBoost processes efficiently without creating thousands of dummy variables
- Prevents overfitting - built-in regularization and early stopping ensure reliable predictions on new data
- Interpretable results - clearly shows location features drive 55% of prediction accuracy



Top 15 features ranked by CatBoost importance scores. Geographic features (City, County, Latitude, Longitude) account for 55% of predictive power.

CatBoost



Metrics:

- R^2 Score: 0.8887 (88.9%)
- Mean Absolute Error: \$161,226
- RMSE: \$322,531
- Dataset: 88,920 properties

What this means:

- Model explains 89% of variance in house prices
- Average prediction error: ~\$161K
- Half of predictions within 8% of actual price (median performance)
- Strong generalization: testing
- Handles price range from \$190K to \$8.5M effectively
- Unbiased predictions - no systematic over/under-estimating
- Performs consistently across all price ranges except ultra-luxury (\$5M+)

Model Comparison

	Ridge/Lasso	Random Forest	XGBoost	LightGBM	CatBoost
Pros/Cons	Pro: Prevents overfitting by shrinking coefficients for stable models Con: Struggles with complex nonlinear relationships	Pro: Handles nonlinearities well and is robust to outliers Con: Can be slow and heavy with many trees	Pro: Strong accuracy with powerful regularization and optimization Con: Training is typically slower on large datasets due to its algorithmic complexity	Pro: Fast and accurate on large, high-dimensional datasets. Con: More sensitive to and risk of overfitting	Pro: Handles categorical features extremely well with little to no preprocessing Con: Training can be slower due to its more complex algorithms
R ² (explains how much variance the model captures)	Training R ² : 0.4896 Testing R ² : 0.4892	Training R ² : 0.8834 Testing R ² : 0.88	Training R ² : 0.88 Testing R ² : 0.85	Training R ² : 0.988304 Testing R ² : 0.871841	Training R ² : 0.9391 Testing R ² : 0.8887
MAPE (average % error between predictions and actuals)	Training MAPE: 40.84 Testing MAPE: 40.83	Training MAPE: 13.78% Testing MAPE: 13.90%	Training MAPE: 16.35% Testing MAPE: 16.98%	Training MAPE: 7.094309 Testing MAPE: 15.242247	Training MAPE: 9.06% Testing MAPE: 11.60%
MdAPE (median % error)	Training MdAPE: 28.18 Testing MdAPE: 28.23	Training MdAPE: 13.24% Testing MdAPE: 11.78	Training MdAPE: 11.65% Testing MdAPE: 11.35%	Training MdAPE: 5.080679 Testing MdAPE: 11.053140	Training MdAPE: 6.57% Testing MdAPE: 8.03%

Next Steps



Thank you!

